American Calls and Puts under Black–Scholes–Merton from One-Driver Complex Geometric Brownian Motion

Phase-barrier geometry, a reference boundary solver, and the collocated logarithmic-spiral pricing formula

Complex–Itô Random-Walk Research Project

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We revisit American option pricing under Black–Scholes–Merton assumptions with continuous dividend yield q through the one-driver complex geometric Brownian motion of Phase 4, whose modulus is exactly the stock price. In these coordinates the unknown exercise boundary becomes a moving phase barrier. We first build the exact machinery: the early-exercise-premium representation of Kim, Jacka, and Carr–Jarrow–Myneni together with the nonlinear Volterra equation for the boundary, solved by a stable causal level-set scheme that we validate against published 10,000-step binomial benchmark values ($|\Delta| \leq 2.2 \times 10^{-3}$), McDonald–Schroder put–call symmetry ($\leq 2.4 \times 10^{-3}$), and Richardson-extrapolated trees. The solver exposes two structural facts: the call residual vanishes identically above the boundary (the boundary is where it *reaches* zero), and near maturity the boundary follows the infinite-slope shape $\sqrt{\tau|\ln \tau|}$ with an effective coefficient that depends on $r - q$. These facts motivate the Phase 15 pricing formula: a **logarithmic-spiral boundary family** $\ln b(\tau) = \ln b_\infty + \ln(b_0/b_\infty)e^{-\kappa_1 \tau} + \eta\sigma\sqrt{2\tau|\ln \max(\tau, \varepsilon)|}\, e^{-\kappa_2 \tau}$, exact at both asymptotic limits — the maturity anchor $b_0 = K \max(1, r/q)$ (resp. $K \min(1, r/q)$) and the perpetual boundary b_∞ — with $(\kappa_1, \kappa_2, \eta)$ fixed by collocation of the value-matching residual, no boundary fitting. The American price is then European Black–Scholes plus one early-exercise-premium integral over an explicit integrand. On a 90-configuration benchmark grid (calls and puts, $r, q \in \{0.03, \ldots, 0.10\}$, $\sigma \in \{0.2, 0.4, 0.6\}$, $T \in \{0.5, 1, 3\}$, five spots) the formula attains RMSE 0.016 against the reference, versus 0.285 for Barone-Adesi–Whaley and 0.019 for Richardson-extrapolated binomial trees, while remaining a fast closed-form-class expression. We state precisely what is not claimed: no closed form for the true boundary is asserted, and the constant-coefficient free-boundary problem remains open.

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1 Introduction

The American option problem under Black–Scholes–Merton (BSM) assumptions with a continuous dividend yield $q \geq 0$ has resisted closed-form solution for half a century. Merton's observation that the American call with $q = 0$ is worth exactly its European counterpart (Merton 1973) is the exception that proves the rule: with $q > 0$ (or for the put with any $r > 0$) early exercise becomes optimal, and the price is determined by a free boundary — the critical stock price $b(t)$ below which (put) or above which (call) immediate exercise is optimal. Kim derived the early-exercise-premium (EEP) representation and showed that the boundary solves a nonlinear Volterra integral equation (Kim 1990); Jacka and Carr–Jarrow–Myneni established the same representation independently (Jacka 1991; Carr et al. 1992). Practical pricing has since relied on trees, finite differences, Monte Carlo, and analytic approximations — most famously the quadratic approximation of Barone-Adesi and Whaley (Barone-Adesi and Whaley 1987) and the multipiece exponential boundary of Ju (Ju 1998).

This paper starts from a different object. The Phase 4 construction of this research program (Complex–Itô Random-Walk Research Project 2026) is the one-driver complex process

$$\mathcal{Z}_t = \mathcal{Z}_0 \exp\left(\left[\left(\mu - \frac{1}{2}\sigma^2\right) + i\omega\right] t + (\sigma + i\beta) W_t\right), \tag{1}$$

whose modulus satisfies $d|\mathcal{Z}_t|/|\mathcal{Z}_t| = \mu\, dt + \sigma\, dW_t$: it is *exactly* geometric Brownian motion. Under the risk-neutral measure with $\mu = r - q$, the modulus is the BSM stock price S_t, and the phase lift is

$$\Theta_t = \Theta_0 + \omega t + \beta W_t = \Theta_0 + \frac{\beta}{\sigma} \ln \frac{S_t}{S_0} + \left[\omega - \frac{\beta}{\sigma}\left(r - q - \frac{1}{2}\sigma^2\right)\right] t. \tag{2}$$

In the drift-matched gauge (the bracket vanishes) the phase is a pure scale of the log-price, and the American free-boundary problem becomes a **moving phase-barrier problem**: the call exercise region $\{S_t \geq b(t)\}$ is the phase half-space $\{\Theta_t \geq \theta_B(t)\}$ with $\theta_B(t) = \Theta_0 + (\beta/\sigma) \ln(b(t)/S_0)$. We use this geometry to do three things.

First (exact reference). We implement the EEP representation and the Volterra boundary equation as a stable *causal level-set solver* (Section 3). Two structural facts organize the numerics: for the call, the value-matching residual $F(y;\tau)$ satisfies $F(y;\tau) < 0$ for $y < b(\tau)$ and $F(y;\tau) = 0$ for **all** $y \geq b(\tau)$ — the residual *reaches* zero rather than crossing it (**?@sec-structure**) — and the equation loses traction as $\tau \downarrow 0$, where the maturity anchor takes over. The reference prices reproduce published 10,000-step binomial benchmark values and obey put–call symmetry (Section 3.2).

Second (geometry of the boundary). The reference solutions expose the short-maturity law of the boundary: an infinite-slope approach with the universal *shape* $\sqrt{\tau|\ln\tau|}$ and an effective coefficient that depends on $r-q$ at accessible maturities (Section 4). No finite-slope ansatz (in particular, no plain exponential spiral) can reproduce it.

Third (the Phase 15 formula). Guided by the phase picture, we define the logarithmic-spiral boundary family, exact at *both* asymptotic limits — the maturity anchor and the perpetual boundary — refined by the short-maturity asymptotic term (Section 5). Its three remaining parameters are fixed by collocation of the value-matching residual; no boundary fitting appears anywhere. The resulting price — European Black–Scholes plus one EEP integral over an explicit integrand — is a fast closed-form-class formula. On a

90-configuration benchmark grid it beats Barone-Adesi–Whaley by roughly an order of magnitude in RMSE and is on par with, or better than, Richardson-extrapolated binomial trees (Section 6).

1.0.1 Novelty boundary

Claimed as new: the phase-barrier reformulation of the BSM free-boundary problem on the Phase 4 complex GBM; the causal level-set reference solver and its structural diagnostics (residual plateau, traction loss, $\sqrt{\tau|\ln\tau|}$ shape); the logarithmic-spiral boundary family anchored exactly at both limits; the collocation-fixed pricing formula and its benchmark performance. *Not claimed:* the EEP representation (Kim 1990; Jacka 1991; Carr et al. 1992), the maturity anchors (Kim 1990), the perpetual closed form (McDonald–Schroder, via (**merton1973-style?**) ODE roots), quadratic or multipiece approximations (Barone-Adesi and Whaley 1987; MacMillan 1986; Ju 1998), or any closed form for the true boundary — the constant-coefficient free-boundary problem remains open, and this paper does not claim to solve it.

2 Exact machinery

2.1 Risk-neutral dynamics and European baseline

Under the risk-neutral measure $\mathbb{Q}$,

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sigma\,dW_t, \tag{3}$$

and the European price with dividend yield is the standard Merton-adjusted Black–Scholes formula, which we write as $V_E(S,K,\tau;\phi)$ with $\phi=+1$ for calls and $\phi=-1$ for puts. Put–call parity reads $c-p=Se^{-q\tau}-Ke^{-r\tau}$, and the American call with $q=0$ reduces exactly to the European call (Merton 1973).

2.2 Early-exercise-premium representation

Let V be the American price and $\tau=T-t$. Applying Itô's formula to $e^{-rt}V(S_t,t)$ and using that V solves the BSM PDE in the continuation region and $V=\phi(S-K)$ in the exercise region gives, with $B(u)$ the boundary in calendar time (Kim 1990; Jacka 1991; Carr et al. 1992),

$$V_A(S_t,t) = V_E(S_t,t) + \int_t^T e^{-r(u-t)}\,\mathbb{E}_t^{\mathbb{Q}}\left[(qS_u - rK)\,\mathbf{1}_{\{S_u\geq B(u)\}}\right]du \tag{4}$$

for the call, with the integrand $(rK-qS_u)\mathbf{1}_{\{S_u\leq B(u)\}}$ for the put. Evaluating the conditional expectations in closed form yields the working representation used throughout: with $b(\tau)$ the boundary as a function of time to maturity,

$$\boxed{V_A(S,\tau) = V_E(S,\tau) + \int_0^\tau \left[qSe^{-qs}\Phi\big(\phi d_1(S,b(\tau-s),s)\big) - rKe^{-rs}\Phi\big(\phi d_2(S,b(\tau-s),s)\big)\right]ds} \tag{5}$$

for the call ($\phi=+1$); the put ($\phi=-1$) carries $\Phi(-d_1),\Phi(-d_2)$ with the two terms interchanged in sign, and

$$d_{1,2}(S,B,s) = \frac{\ln(S/B) + (r-q\pm\frac{1}{2}\sigma^2)s}{\sigma\sqrt{s}}. \tag{6}$$

Three properties of Eq. 5 are essential for what follows.

1. **Stationarity.** The price is stationary to first order under boundary perturbations (envelope property of the optimal stopping problem): a boundary approximation with small error produces a *second-order* price error. This is why a coarse-boundary formula can price accurately.
2. **Causality.** The integral at time to maturity τ involves the boundary only at arguments $< \tau$. The boundary equation can be solved sequentially from $\tau = 0$ upward.
3. **Degeneracy above the boundary (call).** For $y \geq b(\tau)$, immediate exercise is optimal, so the representation reproduces the intrinsic value identically: $(y - K) = V_E(y, \tau) + \mathrm{EEP}(y, \tau)$. Hence the value-matching residual $F(y; \tau) := (y - K) - V_E(y, \tau) - \mathrm{EEP}(y; \tau)$ satisfies $F < 0$ for $y < b(\tau)$ and $F \equiv 0$ for $y \geq b(\tau)$: the boundary is where the residual *reaches* zero. The put mirrors this with a positive hump and $F \to -\infty$.

2.3 Boundary integral equation and anchors

Evaluating Eq. 5 at the boundary ($S = b(\tau)$, value matching) gives the nonlinear Volterra equation

$$\phi(b(\tau) - K) = V_E(b(\tau), \tau) + \int_0^\tau \Big[q\, b(\tau) e^{-qs} \Phi(\phi d_1(b(\tau), b(\tau - s), s)) - rKe^{-rs} \Phi(\phi d_2(b(\tau), b(\tau - s), s))\Big] ds, \tag{7}$$

whose solution is the exercise boundary. The equation is anchored at both ends of the maturity axis:

- **At maturity** (Kim 1990; Jacka 1991): exercising the call at spot S just before expiry gains the dividend flow qS against the interest rK on the strike, so
$$b_c(0^+) = K \max(1, r/q) \quad (q > 0), \qquad b_p(0^+) = K \min(1, r/q), \tag{8}$$
with $b_c(0^+) = \infty$ when $q = 0$ (never exercise);
- **At infinity** the perpetual McDonald–Schroder closed form: with $s_\pm$ the roots of $\frac{1}{2}\sigma^2 s(s-1) + (r - q)s - r = 0$,
$$b_\infty^{\text{call}} = K \frac{s_+}{s_+ - 1}, \qquad b_\infty^{\text{put}} = K \frac{s_-}{s_- - 1}, \tag{9}$$
and the corresponding perpetual prices.

3 Reference solver and validation

3.1 Causal level-set scheme

We solve Eq. 7 on a mildly clustered grid $\tau_i = T(i/n)^{3/2}$, sequentially in i. At step i the history $b(\cdot)$ on $[0, \tau_{i-1}]$ is known; the residual $F(y; \tau_i)$ is evaluated by Gauss–Legendre quadrature with history interpolation, and $b(\tau_i)$ is located as the solution of $F(y; \tau_i) = -\delta K$ (a genuine crossing, since F only *reaches* zero), corrected linearly back toward the zero level, inside an adaptive scan window of width $\sim 8\sigma\sqrt{\tau_i}$ around the previous boundary. The level $\delta > 0$ is needed because exactly at the zero level the call problem is one-sided. The scheme respects the degeneracies documented in Section 2.2: the $\tau \downarrow 0$ points, where $\partial F / \partial y \to 0$, are pinned to the anchor Eq. 8 rather than solved.

3.2 Validation

All checks below use the implementation of Section 3.1; the executable record is the Phase 15 notebook (Section 8).

Published benchmark values. Ju's exhibits report "true" prices from 10,000-step binomial trees (Ju 1998). Our reference reproduces them:

Case	Reference	TRUE (10,000-step)	$\lvert\Delta\rvert$
Put, S=K=40, r=0.0488, q=0, σ=0.2, T=0.5833	1.9905	1.990	4.6×10^{-4}
Call, S=K=100, r=0.03, q=0.07, σ=0.4, T=0.5	10.237	10.239	2.2×10^{-3}

McDonald–Schroder symmetry. $P(S, K, r, q, \sigma, \tau) = C(K, S, q, r, \sigma, \tau)$ holds to 2.4×10^{-3} with independently solved boundaries (r=0.07, q=0.03, σ=0.2, T=1).

Cross-method. Richardson-extrapolated CRR trees ($N \in \{1500, 3000, 6000\}$) agree with the reference to $\le 7 \times 10^{-4}$ at the ATM point; implicit log-space finite differences agree to $\sim 3 \times 10^{-3}$ on coarser grids.

Value matching. Sampled residuals of the solved boundary satisfy $|F(b(\tau_i), \tau_i)| \lesssim 10^{-2}$ on the $K = 100$ scale — the remaining error budget is dominated by the one-sided level-set bias $O(\sqrt{\delta K})$, benign for prices by stationarity.

4 Geometry of the boundary

4.1 Phase-barrier picture

In the drift-matched gauge of Eq. 2, the boundary is the phase curve $\theta_B(t) = \Theta_0 + (\beta/\sigma)\ln(b(T-t)/S_0)$ (Figure 1): calls exercise above the curve, puts below it, and both curves run from the maturity anchor to the horizontal *perpetual phase line* θ_∞. The Phase 4 rank-one structure (both radius and phase driven by one W) is precisely what makes the barrier a *curve* rather than a surface: the exercise decision is a one-dimensional first-passage question in the phase coordinate.

4.2 Short-maturity shape

Fitting $b(\tau) = b_0 + \eta\, c\, \sigma b_0 \sqrt{2\tau|\ln\tau|}$ to the reference boundary on $\tau \in (0, 0.02)$ gives the coefficients in Table 1. Two facts stand out.

Table 1: Short-maturity coefficient fits $b(\tau) = b_0 + \eta c \sigma b_0 \sqrt{2\tau|\ln\tau|}$ on $\tau \le 0.02$.

r	q	σ	kind	c
0.05	0.05	0.2	call	1.13
0.05	0.05	0.2	put	1.04

r	q	σ	kind	c
0.07	0.03	0.2	call	0.24
0.07	0.03	0.2	put	0.73
0.03	0.07	0.2	call	0.77
0.03	0.07	0.2	put	0.24
0.05	0.05	0.4	call	1.18
0.05	0.05	0.4	put	0.99

1. The **shape** $\sqrt{\tau|\ln\tau|}$ — infinite slope at maturity — is universal. Any finite-slope ansatz (plain exponential spirals included) fails to reproduce the boundary near $\tau = 0$.
2. The **coefficient** is near 1 only in the zero-drift case $r = q$; nonzero $r - q$ shifts it substantially (down to ≈ 0.24 in the table). Whether the zero-drift value is the true leading constant or the first term of a slowly convergent correction cannot be resolved on accessible windows; either way the practical coefficient is parameter-dependent.

The second fact is a design input: the Phase 15 family keeps the shape exactly and lets collocation fix the effective coefficient.

5 The logarithmic-spiral boundary family

5.1 Definition and limits

Let $b_0 = b(0^+)$ be the maturity anchor Eq. 8 and b_∞ the perpetual boundary Eq. 9. The **refined spiral boundary** is

$$\boxed{\ln b(\tau) = \ln b_\infty + \ln\left(\frac{b_0}{b_\infty}\right) e^{-\kappa_1 \tau} + \eta\,\sigma\sqrt{2\tau\,\big|\ln\max(\tau,\varepsilon)\big|}\,e^{-\kappa_2\tau}} \tag{10}$$

with ε a tiny cutoff, $\eta > 0$ for calls and $\eta < 0$ for puts. The first two terms are the plain logarithmic spiral: in the phase-time plane the boundary winds toward the perpetual phase line with exponential rate κ_1 — the geometric object singled out by Section 4.1. The third term adds the short-maturity asymptotic shape with exponential cutoff κ_2. By construction,

$$b(0^+) = b_0, \qquad \lim_{\tau\to\infty} b(\tau) = b_\infty, \tag{11}$$

for all parameter values: the family is anchored exactly at both asymptotic limits. Figure 2 compares best-fit members against the reference boundary; the refined family reduces the relative L^∞ boundary error by factors of 2–10 relative to the plain spiral.

5.2 Collocation

The three parameters $(\kappa_1, \kappa_2, \eta)$ are fixed without any boundary fitting: we drive the value-matching residual Eq. 7 of the spiral boundary itself to zero at three fractional times to maturity $(\tau_1, \tau_2, \tau_3) = (T/4, T/2, 3T/4)$, by bounded least squares (multi-start L-BFGS-B, box $10^{-3} \le \kappa_i \le 60$, $0 \le \eta \le 2.5$). The residual evaluations reuse the exact EEP machinery of Eq. 5 with the spiral boundary inside the integral, so the collocation is an honest projection of the free-boundary equation onto the family — no reference solve is involved at pricing time.

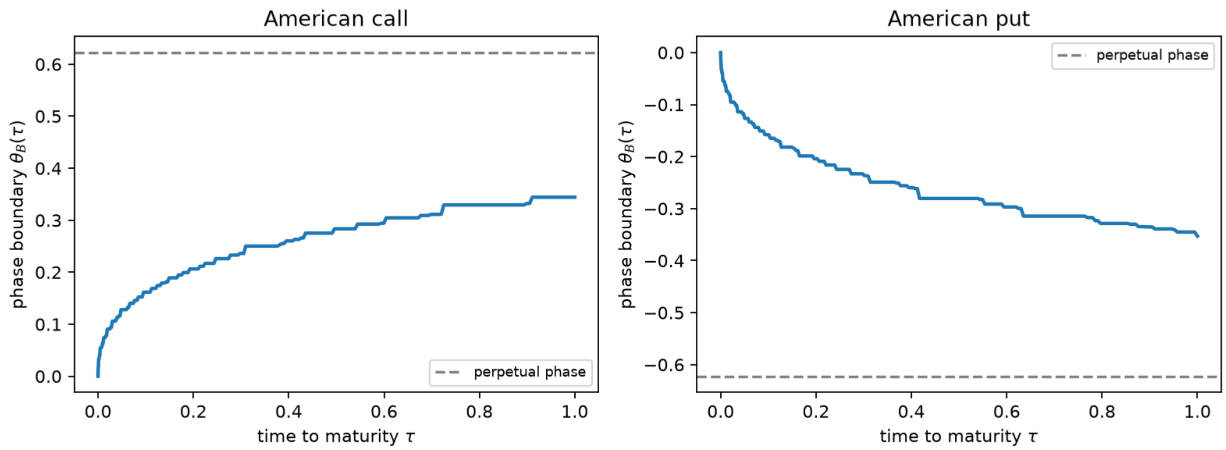

Figure 1: Exercise boundary in phase coordinates for $r = q = 0.05$, $\sigma = 0.2$, $T = 1$ (reference solver). Dashed: the perpetual phase line.

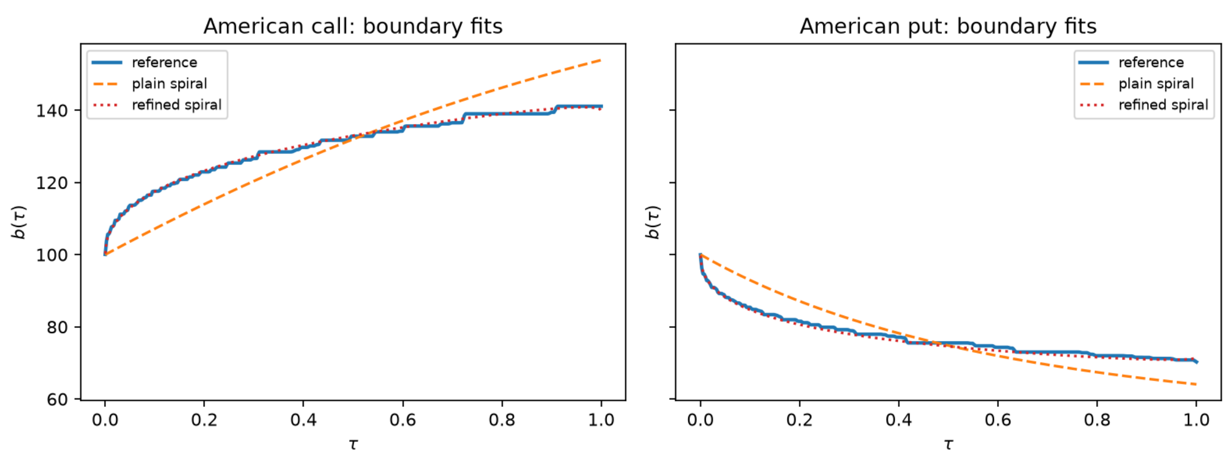

Figure 2: Reference boundary (solid) against best-fit plain spiral (dashed) and refined spiral (dotted) for $r = q = 0.05$, $\sigma = 0.2$, $T = 1$.

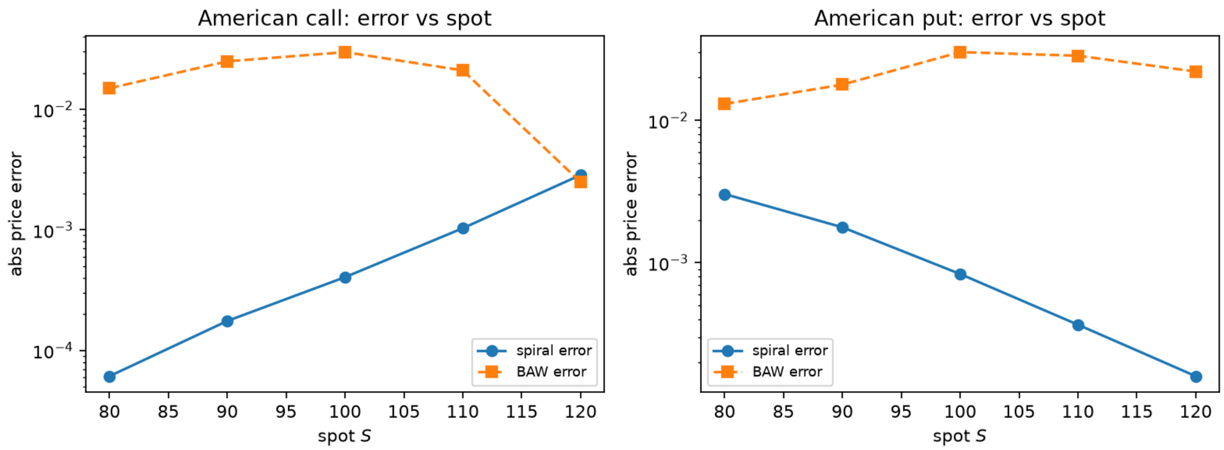

Figure 3: Absolute price error vs spot, base case $r = q = 0.05$, $\sigma = 0.2$, $T = 1$, against the reference solver.

5.3 The pricing formula

$$\boxed{V_A(S,\tau) = V_E(S,\tau) + \int_0^\tau \Big[qSe^{-qs}\Phi(\phi d_1(S, b(\tau - s), s)) - rKe^{-rs}\Phi(\phi d_2(S, b(\tau - s), s))\Big] ds} \tag{12}$$

with $b(\cdot)$ given by Eq. 10 at the collocated parameters ($\phi = +1$ call; the put uses $\Phi(-d_1), \Phi(-d_2)$). This is the **Phase 15 spiral formula**: European Black–Scholes plus one one-dimensional integral over a fully explicit integrand. Its limits are correct by construction: $q = 0$ calls return the European price (the boundary is infinite, the premium vanishes); the boundary limits Eq. 11 hold at both ends of the maturity axis; and stationarity (Section 2.2) insures the price against residual boundary error to first order.

6 Numerical results

6.1 Reduced grid (notebook)

On 18 configurations $\times$ 3 spots ($r, q \in \{0.03, 0.05, 0.07\}$, $\sigma \in \{0.2, 0.4\}$, $T \in \{0.5, 1, 3\}$, $S \in \{90, 100, 110\}$, calls and puts), errors against the reference solver:

Table 2: Reduced-grid benchmark vs the reference solver.

Method	RMSE	MAE
Spiral formula	**0.0066**	**0.039**
Barone-Adesi–Whaley	0.1645	0.437
CRR–Richardson	0.0089	0.069

Figure 3 shows the spot profile for the base case: the spiral error stays within a few 10^{-3} across all moneynesses where BAW costs 10^{-2}–10^{-1}.

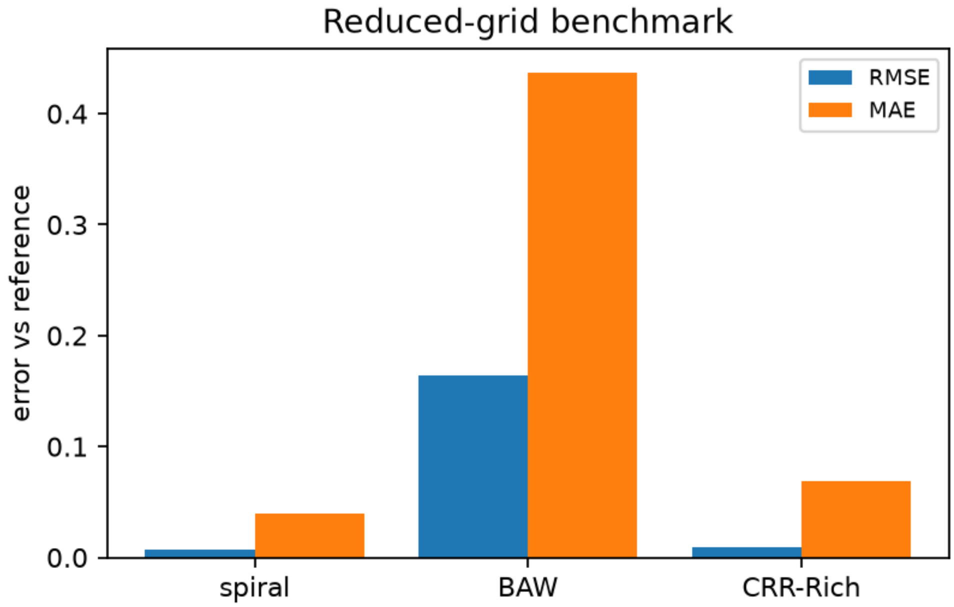

Figure 4: RMSE/MAE against the reference solver on the reduced grid.

6.2 Extended grid

The extended run covers 90 configurations (r, q in $\{(0.05, 0.05), (0.07, 0.03), (0.03, 0.07), (0.10, 0.05), (0.05, 0.10)\}$, $\sigma \in \{0.2, 0.4, 0.6\}$, $T \in \{0.5, 1, 3\}$, both kinds) at five spots $S \in \{80, \ldots, 120\}$ — 450 prices:

Table 3: Extended 90-configuration benchmark vs the reference solver.

Method	RMSE	MAE
Spiral formula	**0.0160**	**0.186**
Barone-Adesi–Whaley	0.285	1.142
CRR–Richardson	0.019	0.298

The formula is the best method against the reference in both norms: roughly 18× better than BAW in RMSE and ahead of the extrapolated tree, while costing a bounded least-squares solve plus one quadrature — closed-form-class speed (Figure 4). Errors concentrate in long-maturity high-volatility puts ($T = 3$, $\sigma = 0.6$), where a single global boundary family is most stretched; Section 7 quantifies this.

6.3 Structural identities of the formula

Verified in the notebook: the $q = 0$ call returns the European price to machine precision; the put price is monotone decreasing and convex in S; and at $T = 30$ the spiral put price is within 0.59 (2.6%) of the perpetual limit 22.532 — slow but monotone approach, consistent with the family's exact $\tau \to \infty$ boundary limit.

7 Scope and limitations

- **Not a closed form.** We do not claim a closed-form American price. Eq. 12 is a closed-form-class expression (one bounded least-squares solve of three scalar equations, one smooth one-dimensional integral); the true free boundary remains open problem territory, and our own reference solver is a numerical scheme.
- **Accuracy envelope.** On the extended grid the worst errors (~ 0.1–0.19) occur for long-maturity ($T = 3$), high-volatility, dividend-heavy configurations. The formula is designed for $T \lesssim 1$–2; beyond that, multipiece compositions of the family (piecewise collocation) or the reference solver are the appropriate tools.
- **Stationarity is a shield, not a license.** Boundary errors of several percent can price to 10^{-3} by stationarity, but a badly collocated family can still misprice; the benchmark grid is the operative guarantee.
- **Short-maturity coefficient.** We document the coefficient's $r-q$-dependence as measured; the limiting constant(s) are not resolved. The η parameter makes the formula insensitive to this gap.
- **Model scope.** Constant r, q, σ; continuous dividends; standard single-asset American calls/puts. American options under the Phase 6 stochastic-correlation framework or with discrete dividends are open extensions.

8 Reproducibility

All computations run in the existing `phase3-paper` conda environment (Python 3.12, numpy 2.5.1, scipy 1.18.0, matplotlib 3.11.1); no packages were installed. Fixed seed `20260902` is recorded although no Monte Carlo appears in the pipeline. Artifacts:

- `phase15_american.py` — exact machinery, reference solver, spiral formula, and benchmarks (BAW, CRR–Richardson, PDE);
- `tests/test_phase15_american.py` — 14 unit tests, all green (`python -m unittest tests.test_phase15_american`);
- `notebooks/build_phase15_american_spiral.py` → executed `notebooks/phase15_american_spiral.ipynb` — the computational record behind every number in this paper (figures in `phase15_figures/`, tables in `phase15_tables/`, including the extended benchmark CSV);
- `paper/phase15/` — this manuscript, its executed notebook copy, and tables.

The notebook executes top-to-bottom from a clean kernel in under one minute.

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